

Cuk converter

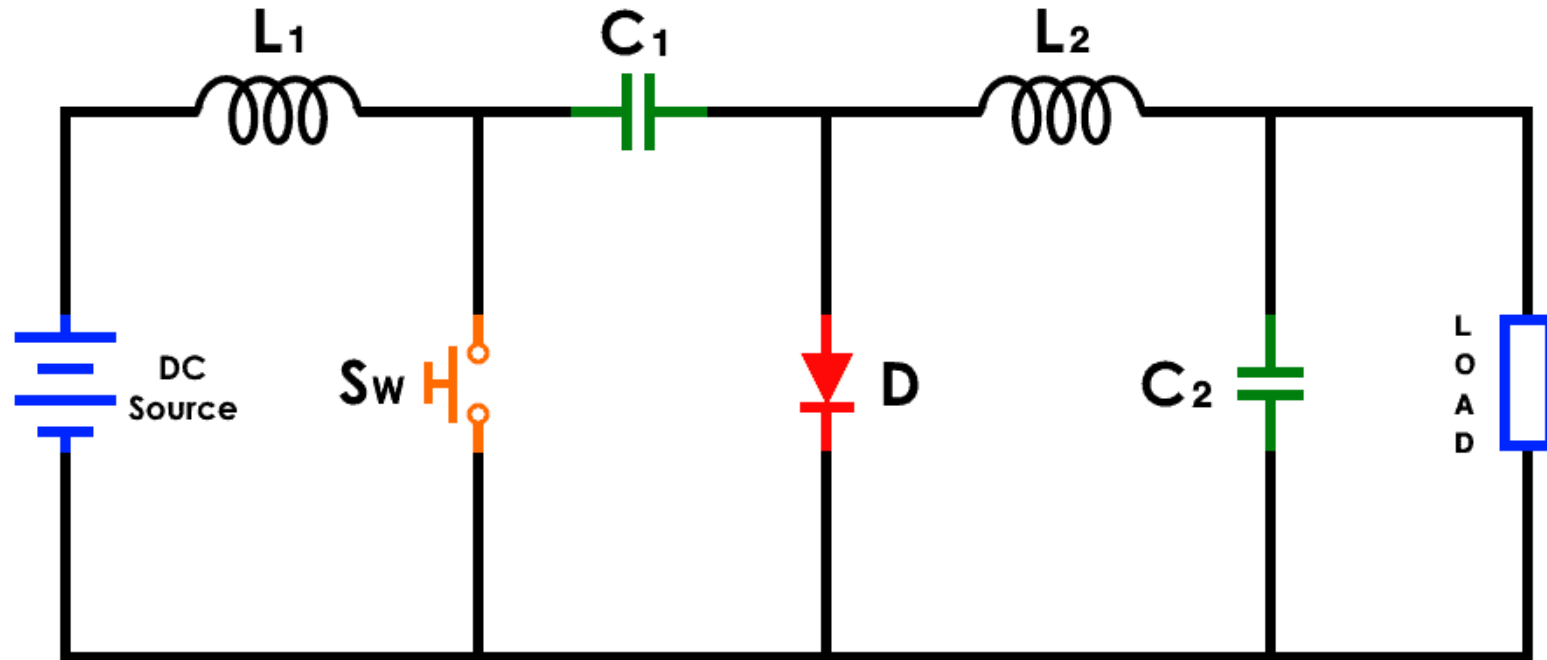
(LAB)

What is a Cuk Converter?

The Cuk converter is a DC- DC-based power electronic converter named after Slobodan Ćuk as its Inventor. The Cuk converter was designed to overcome the drawbacks observed in the buck-boost converter.

The Cuk converter works with the same duality principle where the output of the Cuk converter will reverse polarity similar to the buck-boost converter.

The traditional DC-DC converter buck, boost, and buck-boost converter have one capacitor, one inductor, one diode, and one semiconductor switch. The Cuk converter consists of two inductors, two capacitors, one diode, and one semiconductor switch. The use of two inductors in the Cuk converter topology has a great advantage such that the voltage ripple is less at the input and output side. In this section, we will learn in detail the Cuk converter topology.



Advantages

1. Continuous flow of input and Output current, Energy is supplied to the Load even when the switch is turned OFF.
2. The Input and Output voltage ripple is less in this Cuk converter topology due to inductor L_1 on the input side and inductor L_2 on the output side.
3. We can step up and step down the output voltage in this topology. If the duty ratio for this topology is less than 0.5 then it will act as a step-down converter and if the duty ratio is more than 0.5 then it will act as a step-up converter
4. Can minimize the component size by making inductor L_1 and L_2 wound on single-core
5. High efficiency compared to buck-boost topology

Disadvantages

1. Increase in component count compared to buck-boost topology
2. High current stress on the Switch due to dynamic operation
3. Capacitor C_1 is of large size which acts as the voltage source when the switch is turned off
4. Requirement for inverting mechanism for output voltage. Since the output voltage is reversed as compared to the input voltage
5. Have smooth current waveform due to inductor and capacitor at both input and output sides.

Applications of Cuk converter

1. Use in renewable systems as voltage regulator in hybrid solar wind technology where input voltage depends on sun and wind. It adjusts the output voltage as per inputs in a variation of solar and wind intensity.
2. Use in the battery management system of electric vehicle technology.
3. Use when lesser or greater magnitude of output voltage is needed as compared to the input voltage.
4. Use in low standby continues and simultaneous current and mostly in high and negative polarity output voltage levels.

```
1 vs=12;
2 v0=-18;
3 ro=0.01;
4 rc1=0.05;
5 f=50000;
6 load=40;
7 d=(-v0/(vs-v0))
8 il1=load/vs
9 il2=load/-v0
10 delta_il1=il1*0.1
11 delta_il2=il2*0.1
12 l1=(vs*d)/(f*delta_il1)
13 l2=(vs*d)/(f*delta_il2)
14 c2=(1-d)/(8*ro*l2*f*f)
15 delta_vc1=(vs-v0)*rc1
16 R=(v0*v0)/load
17 c1=(v0*d)/(R*f*delta_vc1)|
```

Given parameters:

$$V_s = 12 \text{ V}$$

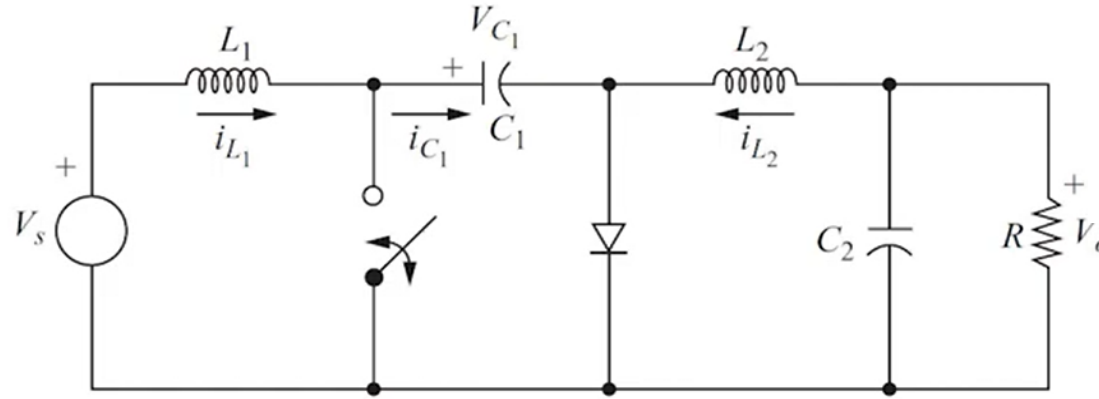
$$r_{C_1} = 5\%$$

$$V_o = -18 \text{ V}$$

$$f = 50000 \text{ Hz}$$

$$r_o = 1\%$$

$$\text{Load} = 40 \text{ W}$$



Boost Buck
dc to dc
converter

1. Calculate duty ratio

$$D = \frac{-V_o}{V_s - V_o} = \frac{-(-18)}{12 - (-18)} = 0.6$$

2. Calculate average inductance current I_{L_1}

$$I_{L_1} = \frac{P_s}{V_s} = \frac{40}{12} = 3.33 \text{ A}$$

3. Calculate average inductance current I_{L_2}

$$I_{L_2} = \frac{P_s}{-V_o} = \frac{40}{-(-18)} = 2.22 \text{ A}$$

4. Calculate rate of change inductor current Δi_{L_1}

$$\Delta i_{L_1} = I_{L_1} \times 10\% = 3.33 \times 10\% = 0.333$$

5. Calculate rate of change inductor current Δi_{L_2}

$$\Delta i_{L_2} = I_{L_2} \times 10\% = 2.22 \times 10\% = 0.222$$

6. Calculate inductance L_1

$$L_1 = \frac{V_s D}{f \Delta i_{L_1}} = \frac{(12)(0.6)}{(50000)(0.333)} = 432 \text{ }\mu\text{H}$$

Given parameters:

$$V_s = 12 \text{ V}$$

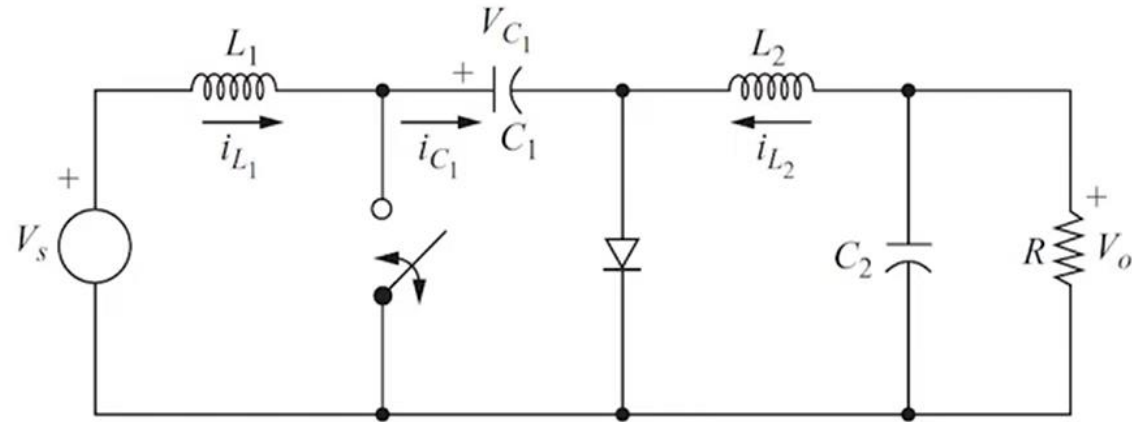
$$r_{C_1} = 5\%$$

$$V_o = -18 \text{ V}$$

$$f = 50000 \text{ Hz}$$

$$r_o = 1\%$$

$$\text{Load} = 40 \text{ W}$$



Boost Buck
dc to dc
converter

7. Calculate inductance L_2

$$L_2 = \frac{V_s D}{f \Delta i_{L_2}} = \frac{(12)(0.6)}{(50000)(0.222)} = \mathbf{648 \text{ } \mu\text{H}}$$

8. Calculate capacitance C_2

$$C_2 = \frac{1 - D}{r 8 L_2 f^2} = 3.08 \text{ } \mu\text{F}$$

9. Calculate rate of change capacitor voltage Δv_{C_1}

$$\Delta v_{C_1} = (V_s - V_o) r_{C_1} = (12 - (-18)) \mathbf{5\%}$$

10. Calculate load resistance

$$R = \frac{V_o^2}{P} = \frac{(18)^2}{40} = 8.1 \text{ } \Omega$$

11. Calculate capacitance C_1

$$C_1 = \frac{V_o D}{R f \Delta v_{C_1}} = \frac{(18)(0.6)}{(8.1)(50000)(1.5)} = 17.8 \text{ } \mu\text{F}$$

